

Notes: Rajan, Servaes & Zingales (JF, 2000)

The Cost of Diversity: The Diversification Discount and Inefficient Investment

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1 Big picture

- **Question.** Why do diversified firms often trade at a discount relative to a portfolio of focused firms, and how does internal capital allocation differ from what stand-alone firms would do in outside capital markets?
- **Core idea.** Internal capital markets are not just about “winner picking.” They also change the *distribution of power* across divisions. Headquarters can move resources across divisions ex ante, but it cannot fully commit to an ex post rule for how the joint surplus will be shared.
- **Main mechanism.** Each division can choose either an **efficient** investment, which maximizes total surplus but leaves the division vulnerable to ex post poaching, or a **defensive** investment, which protects the division’s surplus but produces lower returns. When divisions are very different in resources and opportunities, the strong division may not want to cooperate.
- **The paper’s organizing variable.** The key state variable is **diversity**: dispersion in *resource-weighted opportunities* across a firm’s segments. A segment’s raw opportunities matter, but the theory says what really drives bargaining and transfers is opportunities multiplied by the segment’s size/resources.
- **Main theoretical prediction.** At low diversity, headquarters reallocates resources toward the segment with better opportunities, much like an efficient internal capital market. At higher diversity, headquarters may transfer resources in the **wrong direction**—toward a segment with worse opportunities—because that softens bargaining distortions and keeps the stronger segment willing to make the efficient investment.
- **Empirical setting.** The authors study diversified U.S. firms using COMPUSTAT business segment data over **1979–1993**; the main regressions use **1980–1993** and compare segment investment behavior to single-segment firms in the same 3-digit SIC industry.
- **Baseline allocation fact.** In raw data, high- q segments invest slightly more than low- q segments. But after adjusting for industry investment and for the firm’s average excess investment, **low- q segments receive net transfers and high- q segments make them**. The preferred adjusted-investment measure is about **0.004** for low- q segments and about **-0.002** for high- q segments.
- **Main discriminating result.** Diversity shifts transfers according to **size-weighted opportunities**, not just q . In Table IV, a rise in diversity increases transfers to segments with *below-average weighted opportunities* and decreases transfers from segments with *above-average weighted opportunities*, even when those segments have the same raw q .
- **Value result.** Greater diversity reduces the efficiency of internal allocation and lowers diversified-firm value. A one-standard-deviation increase in diversity lowers excess value by roughly **5 percentage points** in the direct value regressions.
- **Economic takeaway.** Diversity is costly not because diversification is mechanically bad, but because large differences in segment size and opportunities create internal bargaining distortions. Internal capital markets can help when divisions are similar enough; they can hurt when diversity becomes too large.

2 Data and measurement (institutional context and key objects)

2.1 Why diversified segment data are the right laboratory

- The paper wants to compare decisions made *inside hierarchies* with the investment choices stand-alone firms would make in markets.
- Public U.S. firms report segment sales, profitability, investment, and assets by major business line, so segment data provide an unusual window into internal capital budgeting.
- This setting is especially useful because the authors can benchmark each segment against focused firms operating in the same industry.

Interpretation. The empirical design is not “diversified firm versus focused firm” in the aggregate. It is “segment inside a hierarchy” versus “comparable single-segment firms in the same industry.”

2.2 Sample construction and unit of observation

- Segment data come from the COMPUSTAT Business Segment Information files, using both the active and research files.
- The raw segment files contain about **156,598 firm-segment-year observations** over **1979–1993**.
- Because the authors need industry Tobin’s q for comparison firms, they exclude firms with any financial-services segment (SIC starting with 6), since Tobin’s q is harder to construct there.
- Since the comparison q is measured at the beginning of the period, the main usable regression window becomes **1980–1993**.
- The transfer regressions in Table IV use about **13,947 firm-year observations**.

Interpretation. The paper is built on within-firm, over-time variation in diversified firms after matching each segment to a focused-firm industry benchmark.

2.3 Proxy for segment opportunities

- The model’s primitive opportunity parameter is not directly observable.
- The paper proxies for a segment’s opportunities using the **asset-weighted average Tobin’s q of single-segment firms** in the same **3-digit SIC industry**.
- The q ratios are constructed using the Lindenberg–Ross methodology with the Hall et al. implementation.
- To reduce outlier influence, the variables are winsorized at the 1st and 99th percentiles.

Interpretation. The segment's own reported q is not used. Instead, the paper asks: if this segment were a focused firm in the same industry, what opportunities would comparable firms face?

2.4 Proxy for segment resources

- The model's initial resources A_j^0 should capture what a division can deploy before internal transfers.
- The paper uses **beginning-of-year segment assets** as the resource measure, scaled by total firm assets.
- Segment assets are preferred to segment cash flow because segment income can be strategically reported, whereas assets are harder to manipulate and also reflect borrowing capacity.

Interpretation. Resources are about both internally generated funds and capacity to raise funds. Asset share is therefore the paper's empirical proxy for a segment's bargaining weight and internal scale.

2.5 Proxy for transfers: adjusted investment

The paper's best proxy for a transfer is the segment's **firm- and industry-adjusted investment ratio**:

$$Transfer_{jft} = \left(\frac{I_{jft}}{BA_{jft}} - \frac{I_{jt}^{ss}}{BA_{jt}^{ss}} \right) - \sum_{s \in f} w_{sft} \left(\frac{I_{sft}}{BA_{sft}} - \frac{I_{st}^{ss}}{BA_{st}^{ss}} \right), \quad (1)$$

where I is capital expenditure, BA is beginning-of-period book assets, ss denotes the single-segment benchmark in the same industry, and w_{sft} is the segment's asset share within the firm.

Interpretation. The first term asks whether the segment invests more or less than focused firms in the same industry. The second term removes the diversified firm's *overall* tendency to invest more or less than focused firms, so the remaining variation is interpreted as an *internal reallocation* across segments.

2.6 Proxy for relative firm value

The main excess-value measure is Lang–Stulz style:

$$ExcessValue_{ft} = \frac{MV_{ft}}{RVA_{ft}} - \sum_{j \in f} q_{jt}^{ss} \frac{BA_{jft}}{BA_{ft}}, \quad (2)$$

where MV is the market value of assets, RVA is the replacement value of assets, and q_{jt}^{ss} is the asset-weighted average Tobin's q of single-segment firms in segment j 's industry.

The paper also uses an alternative market-to-sales based excess-value measure.

Interpretation. Excess value asks whether the diversified firm is worth more or less than a synthetic portfolio of stand-alone firms operating in the same industries.

2.7 Proxy for diversity

The key explanatory variable is

$$Diversity_{ft} = \frac{\sqrt{\frac{1}{n_f-1} \sum_{j \in f} (w_{jft} q_{jft} - \bar{w} \bar{q}_{ft})^2}}{\bar{q}_{ft}}, \quad (3)$$

where w_{jft} is the segment's asset share, q_{jft} is the benchmark industry q , $\bar{w} \bar{q}_{ft}$ is the firm mean of asset-weighted q , and $\bar{q}_{ft}$ is the equally weighted average segment q .

Interpretation. Diversity is not just dispersion in opportunities and not just dispersion in size. It is dispersion in **size-weighted opportunities**, which is exactly what the model says drives bargaining and cross-subsidization.

2.8 Additional controls

- **Inverse of average q :** $1/\bar{q}$, motivated by the model's closed-form transfer expressions.
- **Firm size:** logarithm of total sales.
- **Herfindahl index of segment size:** a control for firm focus/concentration.
- **Coefficient of variation of segment q :** isolates pure dispersion in opportunities.
- **Coefficient of variation of segment size:** isolates pure dispersion in scale.

Interpretation. These extra variables are there to show that the paper's diversity measure is not just a relabeled focus measure, a pure q -dispersion measure, or a pure size-dispersion measure.

2.9 Descriptive facts

- The average diversity measure is about **0.295**.
- Using the q -based method, diversified firms trade at an average discount of roughly **12 percent**.
- In raw capex-to-assets data, high- q segments invest slightly more than low- q segments: about **0.101** versus **0.096**.
- After industry adjustment, low- q segments receive more funds than high- q segments: about **0.013** versus **0.008**.
- After both industry and firm adjustment, the paper's preferred transfer measure is about **-0.002** for high- q segments and about **0.004** for low- q segments.

Interpretation. Even before the full theory test, the raw descriptive pattern already says something striking: diversified firms do not simply shovel more money toward their best- q segments.

3 Model and empirical specifications (all equations + interpretation)

3.1 Timing and players

- A diversified firm has two divisions, A and B , each led by a self-interested divisional manager.
- Headquarters maximizes total firm surplus.
- Timeline:
 - Date 0: headquarters can transfer resources across divisions.
 - Date 1: each division chooses its investment type.
 - Date 2: divisions bargain over how the total surplus will be split.
 - Date 3: production occurs and payoffs are realized.

If headquarters transfers t from division A to division B , then

$$A_A^1 = A_A^0 - t, \quad A_B^1 = A_B^0 + t. \quad (4)$$

Interpretation. The headquarters can move cash *before* investment, but it cannot write a complete contract over how future surplus will be divided after investments are chosen.

3.2 Technology, defensive investment, and ex post bargaining

- If division j makes the **efficient** investment, gross return per unit is a_j .
- If it makes the **defensive** investment, gross return is $a_j - y$, where $y > 0$ is the return loss from defensiveness.
- If one division makes the defensive investment while the other makes the efficient one, the defensive division can poach from the efficient division. The efficient division then suffers a larger loss $\theta > y$.
- If both make efficient investments, the divisions avoid poaching and split the joint surplus equally.

The paper's core payoff comparison is therefore between an efficient-but-exposed action and a defensive-but-protected action.

Interpretation. Defensive investment is not just low productivity. It is a way for a division to protect itself in the internal political game.

3.3 Incentive compatibility for efficient investment

Suppose divisions j and k are both expected to invest efficiently. Division j will indeed choose the efficient investment if

$$\frac{1}{2} (a_j A_j^1 + a_k A_k^1) \geq (a_j - y) A_j^1. \quad (5)$$

If the same condition holds for both divisions, the efficient-investment outcome is an equilibrium.

Let $\beta_j = a_j - y$. Under the normalization that $\beta_j A_j^1 \geq \beta_k A_k^1$, the condition can be rewritten as

$$y (A_j^1 + A_k^1) \geq \beta_j A_j^1 - \beta_k A_k^1. \quad (6)$$

Interpretation. The right-hand side is the **diversity problem**: one division's resource-weighted opportunities exceed the other's by too much. When that gap is too large, the strong division would rather defend its own turf than cooperate inside the firm.

3.4 Headquarters transfers and the key comparative statics

There are two cases.

Case 1: the receiving division has better opportunities. If $\beta_B > \beta_A$, transfers from A to B go in the “right” direction and continue until the incentive constraint for efficient investment just binds:

$$\beta_B (A_B^0 + t) = \beta_A (A_A^0 - t) + y (A_A^0 + A_B^0). \quad (7)$$

Solving gives

$$t = \frac{\beta_A A_A^0 - \beta_B A_B^0 + y (A_A^0 + A_B^0)}{\beta_A + \beta_B}. \quad (8)$$

Case 2: the sending division has better opportunities. If $\beta_B < \beta_A$, then low diversity still leads to transfer toward the better division. But when diversity is high, headquarters may instead transfer in the **wrong direction**, from A to B , to keep A willing to choose the efficient investment:

$$\beta_A (A_A^0 - t) = \beta_B (A_B^0 + t) + y (A_A^0 + A_B^0). \quad (9)$$

Wrong-way transfer becomes necessary when

$$\beta_A A_A^0 - \beta_B A_B^0 > y (A_A^0 + A_B^0). \quad (10)$$

The transfer is privately worthwhile for headquarters only if the gain in incentives exceeds the allocative loss, which requires

$$y > \frac{\beta_A - \beta_B}{2}. \quad (11)$$

Interpretation. This is the central theoretical result. The headquarters may rationally create allocative inefficiency at the margin because that is the price of avoiding even worse incentive distortion.

3.5 Main empirical predictions

The paper converts the theory into four empirical predictions by grouping segments according to

- whether q_j is above or below the firm's average $\bar{q}$, and
- whether $w_j q_j$ is above or below the firm's average weighted opportunity $\overline{wq}$.

This creates four segment types:

1. high q , high weighted opportunity;
2. high q , low weighted opportunity;
3. low q , high weighted opportunity;
4. low q , low weighted opportunity.

Interpretation. This four-cell design is how the authors separate their theory from simpler efficient-internal-capital-market stories. Their model says *weighted opportunities*, not raw q , determine who gives and who gets.

3.6 Segment-level transfer regression

For each of the four segment groups g , the paper estimates

$$Transfer_{g,ft} = \alpha_f + \lambda_t + \beta_1 \frac{1}{\bar{q}_{ft}} + \beta_2 Diversity_{ft} + \beta_3 \log(Sales_{ft}) + \varepsilon_{g,ft}, \quad (12)$$

where $Transfer_{g,ft}$ is the asset-weighted sum of adjusted investment for all segments in firm f , year t , that belong to group g ; α_f are firm fixed effects; and λ_t are year effects.

Interpretation. Identification comes from asking whether *changes in diversity within the same diversified firm over time* shift funds toward some segment types and away from others.

3.7 Relative value added by allocation

The paper also compresses all internal reallocation into one summary index:

$$VAA_{ft} = \sum_{j \in f} \frac{BA_{jft}}{BA_{ft}} (q_{jft} - \bar{q}_{ft}) Transfer_{jft}. \quad (13)$$

Interpretation. A transfer toward a segment whose opportunities are above the firm's mean raises VAA ; a transfer toward a below-average- q segment lowers it. This is the paper's direct measure of whether internal allocation is helping or hurting.

3.8 Firm-value regressions

The value-added measure is linked to firm value through

$$ExcessValue_{ft} = \alpha_f + \lambda_t + \delta VAA_{ft} + \gamma \log(Sales_{ft}) + \varepsilon_{ft}. \quad (14)$$

The paper also estimates a direct reduced form:

$$ExcessValue_{ft} = \alpha_f + \lambda_t + \phi_1 \frac{1}{\bar{q}_{ft}} + \phi_2 Diversity_{ft} + \phi_3 Herfindahl_{ft} + \phi_4 \log(Sales_{ft}) + \varepsilon_{ft}. \quad (15)$$

Interpretation. If the theory is right, diversity should hurt value both *indirectly* through worse internal allocation and *directly* in the reduced-form excess-value regression.

4 Identification and empirical design (what makes the estimates credible)

4.1 What is identified by the main estimates

- The paper does *not* try to identify a causal treatment effect of diversification versus focus.
- Instead, it identifies whether **within-firm changes in diversity over time** are associated with predictable changes in internal capital allocation and value.
- Firm fixed effects remove time-invariant differences across firms, such as stable organizational culture, segment-reporting habits, or selection into diversification.

Interpretation. This is an important design choice. The authors know diversification itself is endogenous, so they lean on within-firm variation rather than simple cross-sectional comparisons.

4.2 Why the four-group design separates theories

- Efficient internal capital market theories predict funds should flow toward high- q segments.
- Scharfstein–Stein style socialist cross-subsidy stories predict transfers toward low-productivity divisions.
- Rajan–Servaes–Zingales predict something subtler: the sign depends on whether a segment is above or below the firm’s **weighted** opportunity benchmark.

Interpretation. The most revealing cells are the mixed cases:

- **high q , low weighted opportunity** segments are often small but attractive segments;
- **low q , high weighted opportunity** segments are often large but unattractive segments.

The paper’s theory predicts diversity should move funds *toward the first* and *away from the second*. That is exactly the opposite of a naive winner-picking story.

4.3 Why mismeasured q or external-financing stories are not enough

- One alternative is that Tobin's q is noisy and the market rations focused firms incorrectly, so diversified firms merely correct market mistakes.
- This story can explain some low- q segments receiving funds, but it cannot explain the full four-cell pattern in Table IV.
- More importantly, if internal transfers mainly corrected external market mistakes, they should raise value. Table VI instead shows that the paper's "value added by allocation" measure is positively related to firm value, so wrong-way transfers look costly rather than efficiency enhancing.

Interpretation. The paper's logic is: it is not enough to match the transfer pattern; the transfer pattern must also line up with value.

4.4 Robustness to focus, q -dispersion, and size-dispersion

- Adding the **Herfindahl index of segment size** does not eliminate the diversity effect. If anything, the magnitude of the diversity coefficient often rises.
- Controlling for the **coefficient of variation of segment q** leaves the diversity result essentially intact.
- Controlling for the **coefficient of variation of segment size** absorbs some of the diversity effect in two cells, which makes sense because size dispersion is one ingredient of weighted-opportunity dispersion.

Interpretation. Diversity is related to focus and size dispersion, but the data do not look like a pure focus story. The paper's weighted-opportunity measure still has independent explanatory power.

4.5 Fama–MacBeth style robustness

- The authors worry that residuals may be correlated within years, so ordinary standard errors may be too small.
- They therefore run fixed-effects Fama–MacBeth style robustness checks.
- The coefficient estimates are almost unchanged, and the resulting t -statistics remain large (roughly **2.8 to 8**).

Interpretation. The main allocation results are not a fragile artifact of one specific standard-error choice.

4.6 Nonparametric evidence on the individual-rationality constraint

- The theory says wrong-way transfers should disappear if diversity becomes so large that headquarters no longer wants to preserve incentives.

- The paper therefore runs kernel regressions of transfers on diversity for the segment groups where this constraint matters most.
- Over the sample range, the relationship looks monotone rather than kinked in a way that would suggest the individual-rationality constraint is binding frequently.

Interpretation. In the data range they study, the authors do not find strong evidence that firms are routinely past the point where headquarters gives up on incentive-preserving transfers.

4.7 What the paper can and cannot claim

- The paper cannot directly observe lobbying, bargaining, or poaching.
- It also does not claim that all diversification discounts come from this mechanism.
- What it does show is a remarkably specific pattern: **weighted-opportunity diversity predicts the direction of internal transfers, and those transfer patterns are tied to lower value.**

Interpretation. The evidence is stronger than a generic “conglomerates are inefficient” claim, but weaker than a literal causal proof of internal politics.

5 Tables and figures

5.1 Figure 1: Timing of the model

Read.

- Date 0: headquarters reallocates funds.
- Date 1: divisions choose efficient or defensive investment.
- Date 2: surplus is split.
- Date 3: production and payoffs occur.

Economic message. The timing is the whole point. Headquarters can move resources before investment, but the divisions bargain later. That separation creates room for ex ante transfers to be used as a device to manage ex post political conflict.

5.2 Figure 2: Numerical example of diversity, transfers, and value

Read.

- At low diversity, transfers go toward the better division and the diversified firm can trade at a premium.
- As diversity rises, transfers switch direction and start going toward the worse division.

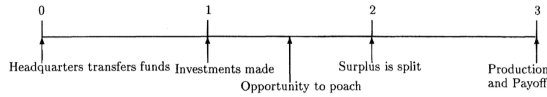


Figure 1. Timing of the events.

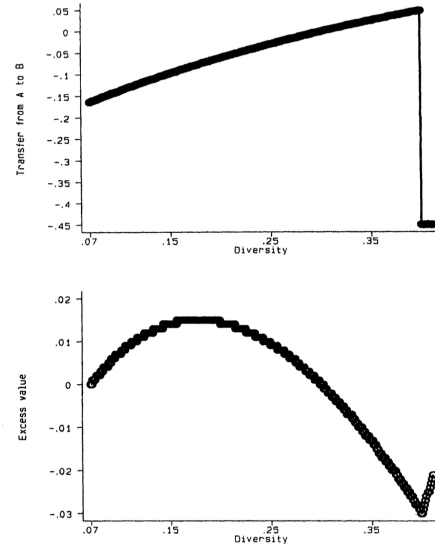


Figure 2. Numerical example of the effects of diversity. This figure presents a numerical example of the effects of diversity on the amount of interdivisional transfers and on the relative value of a diversified firm vis-a-vis a portfolio of single-segment firms. In this example $\lambda_0^H = 0.55$, $\lambda_0^L = 0.45$, $\gamma = 0.3$, $\alpha^H = 1$, and $\alpha^L - \alpha^H$ varies between 0 and 0.62.

- Firm value turns negative and the discount deepens with diversity.
- At extreme diversity, headquarters stops making wrong-way transfers and the economy moves to the third-best outcome.

Economic message. The figure is the model in one picture: diversity first limits efficient winner-picking, then induces inefficient cross-subsidization, and finally becomes so severe that even the second-best internal fix is no longer worth it.

5.3 Table II: Average allocation of funds in diversified firms

Read.

- Raw investment ratio: high- q segments invest about **0.101**; low- q segments about **0.096**.
- Industry-adjusted investment: high- q segments about **0.008**; low- q segments about **0.013**.
- Firm- and industry-adjusted investment: high- q segments about **-0.002**; low- q segments about **0.004**.

Economic message. Once the paper strips out both industry investment needs and the firm's overall investment stance, diversified firms look like they are cross-subsidizing their lower-opportunity segments.

5.4 Table III: Competing theoretical predictions

Read.

- Efficient internal capital market models predict more funds to high- q segments.

Table II
Allocation of Funds in a Diversified Firm

The level of investments in business segments are compared with an industry q above the firm's average and that in business segments with an industry q below the firm's average. We use three definitions of investments: Investment ratio is the capital expenditure to beginning-of-the-period asset ratio, I_j/BA_j , where BA is book value of assets, I is capital expenditures, and subscript j refers to segment j . Industry-adjusted investment ratio is the segment investment ratio less the average industry investment ratio: $(I_j/BA_j) - (I^s/BA^s)$, where (I^s/BA^s) is the asset-weighted capital expenditure to assets ratio for the single-segment firms in the corresponding industry. Firm and industry-adjusted investment ratio is the industry-adjusted investment ratio in a segment less the weighted average industry-adjusted investment ratio across all the segments of the firm. This is defined as

$$\frac{I_j}{BA_j} - \frac{I^s}{BA^s} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I^s}{BA^s} \right),$$

where w_j is the segment's share of total assets. All the data are for the period 1980–1993.

Funds Allocated	Segments with $q > \bar{q}$	Segments with $q < \bar{q}$	Difference
Investment ratio	0.101	0.096	0.005
Industry adjusted	0.008	0.013	−0.005
Firm and industry adjusted	−0.002	0.004	−0.006
Number of segments	23,604	22,600	

Table III

**Effect of Diversity of Opportunities on Internal Transfers:
Theoretical Predictions**

This table summarizes the predictions of the main theories in terms of investments in different segments. Segments are divided according to whether they have better opportunities than the firm's average ($q > \bar{q}$) and more resources-weighted opportunities than the firm's average ($\lambda q > \lambda \bar{q}$). The predictions of "our theory" hold only as long as the headquarters wants to preserve the incentives to make efficient investments.

	Adjusted Investment in Segments with			
	$q > \bar{q}$ $\lambda q > \lambda \bar{q}$	$q > \bar{q}$ $\lambda q < \lambda \bar{q}$	$q < \bar{q}$ $\lambda q > \lambda \bar{q}$	$q < \bar{q}$ $\lambda q < \lambda \bar{q}$
Efficient internal capital market	+	+	−	−
Our theory	−	+	−	+
Scharfstein and Stein	−	−	+	+

- Scharfstein–Stein style models predict more subsidies to the weakest segments.
- Rajan–Servaes–Zingales predict that **size-weighted opportunities** determine the sign.

Economic message. Table III is the map for reading Table IV. It tells the reader exactly which cells distinguish a power/bargaining model from a pure winner-picking model.

5.5 Table IV: Main transfer regressions

Table IV
**Segment Investment and Diversity
in Investment Opportunities**

Firm- and industry-adjusted investment is the industry-adjusted investment in a segment less the weighted average industry-adjusted investments across all the segments of a firm. This is defined as

$$\frac{I_j}{BA_j} - \frac{I^s}{BA^s} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I^s}{BA^s} \right),$$

where w_j is the asset weight of the segment. The inverse of average q equals $1/q\bar{e}$, where $q\bar{e}$ is the equally weighted average q . Diversity is the standard deviation of a firm's asset-weighted q ($\sqrt{\sum_{j=1}^n (w_j q_j - \bar{wq})^2 / (n - 1)}$) divided by the equally weighted average q . Size is the logarithm of total sales. Coefficient of variation of segment qs is the standard deviation of segment qs divided by the mean of segment qs . Coefficient of variation of segment size is the standard deviation of segment shares in a diversified firm divided by mean segment share. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	Adjusted Investment in Segments with			
	$q > \bar{q}$ $\lambda q > \lambda \bar{q}$	$q > \bar{q}$ $\lambda q < \lambda \bar{q}$	$q < \bar{q}$ $\lambda q > \lambda \bar{q}$	$q < \bar{q}$ $\lambda q < \lambda \bar{q}$
Panel A: Basic Specification				
Inverse of average q	0.005 (5.383)	−0.001 (−0.896)	−0.001 (−0.726)	−0.004 (−4.289)
Diversity	−0.014 (−9.059)	0.004 (3.637)	−0.004 (−4.004)	0.014 (9.547)
Firm size	0.000 (−0.192)	0.000 (0.249)	−0.001 (−1.342)	0.001 (1.054)
R^2	0.321	0.332	0.326	0.318
N	13,947	13,947	13,947	13,947
Panel B: The Effect of Focus				
Inverse of average q	0.006 (5.610)	−0.001 (−1.043)	0.000 (−0.600)	−0.004 (−4.506)
Diversity	−0.025 (−12.036)	0.008 (6.743)	−0.009 (−6.786)	0.024 (12.125)
Firm size	0.001 (1.394)	0.000 (−0.794)	0.000 (−0.413)	0.000 (−0.617)
Herfindahl index of segment size	0.025 (9.653)	−0.011 (−6.472)	0.011 (5.633)	−0.023 (−9.602)
R^2	0.329	0.335	0.328	0.326
N	13,947	13,947	13,947	13,947

Table IV—Continued

Panel C: The Effect of the Coefficient of Variation of Segment q				
Inverse of average q	0.005 (4.357)	−0.001 (−1.210)	0.000 (−0.258)	−0.003 (−3.414)
Diversity	−0.015 (−9.316)	0.003 (3.454)	−0.004 (−3.759)	0.015 (9.687)
Coeff. variation of q	−0.006 (−3.797)	−0.002 (−1.554)	0.002 (1.932)	0.005 (3.572)
Firm size	0.000 (0.072)	0.000 (0.353)	−0.001 (−1.475)	0.000 (0.793)
R^2	0.322	0.332	0.326	0.319
N	13,947	13,947	13,947	13,947
Panel D: The Effect of the Coefficient of Variation of Segment Size				
Inverse of average q	0.005 (5.367)	0.000 (−0.592)	−0.001 (−1.052)	−0.004 (−4.247)
Diversity	−0.015 (−8.010)	0.000 (−0.026)	0.000 (−0.055)	0.014 (8.167)
Coeff. variation of segment size	0.000 (0.383)	0.004 (5.685)	−0.005 (−5.961)	0.000 (0.039)
Firm size	0.000 (−0.221)	0.000 (−0.199)	0.000 (−0.852)	0.001 (1.045)
R^2	0.321	0.335	0.328	0.318
N	13,947	13,947	13,947	13,947

Read. In Panel A, the coefficient on diversity is

- **-0.014** for segments with high q and high weighted opportunities;
- **+0.004** for segments with high q but low weighted opportunities;
- **-0.004** for segments with low q but high weighted opportunities;
- **+0.014** for segments with low q and low weighted opportunities.

All four have the sign predicted by the paper’s theory and are statistically significant.

The paper also notes that a one-standard-deviation increase in diversity lowers transfers to high- q , high-weighted-opportunity segments by about **0.0027**, more than 1.5 times the sample mean of that dependent variable.

Economic message. This is the paper’s cleanest evidence. Diversity does not just move money away from high- q segments and toward low- q segments. It moves money according to *weighted* opportunities, exactly as the bargaining model predicts.

5.6 Figure 3: Kernel regressions of transfer on diversity

Read.

- For the segment group that should lose funds when diversity rises, fitted transfers fall with diversity.
- For the group that should receive funds when diversity rises, fitted transfers rise with diversity.
- The plots do not show a strong empirical region where the individual-rationality constraint sharply shuts the mechanism down.

Economic message. The parametric results are not being driven by a single linearity assumption. The data look broadly consistent with the comparative statics the model emphasizes.

5.7 Table V: Value added by allocation falls with diversity

Read.

- In the regression using the relative value-added-by-allocation measure, the diversity coefficient is about **-0.008**.
- In the regression using the absolute value-added-by-allocation measure, the diversity coefficient is about **-0.010**.

Both are statistically significant.

Economic message. Once all segment transfers are aggregated into one firm-level measure of allocative efficiency, diversity still looks bad. The problem is not just who gives and who gets; the overall pattern of reallocation becomes less value enhancing as diversity rises.

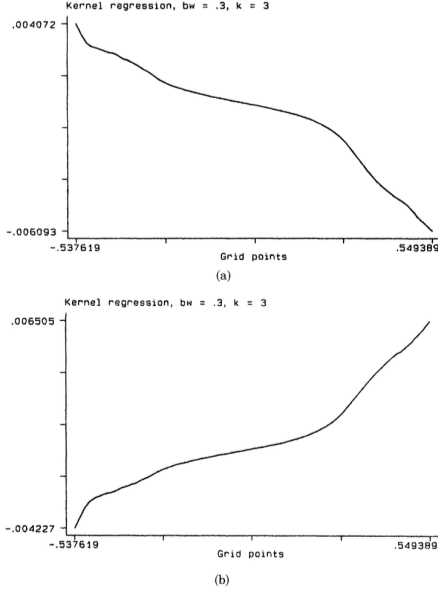


Figure 3. Kernel regression of adjusted investment on diversity. Plots of the fitted values from the kernel regression of adjusted investments against diversity. Before estimating the relationship, we partial out the inverse of the equally weighted q , firm size (logarithm of total sales), firm-specific effects as well as the calendar-year dummies. The kernel density is estimated using the Epanechnikov kernel with a bandwidth of 0.3 and a grid of 100 points. The y-axis contains the fitted values from the kernel regression and the x-axis the corresponding grid points. The top plot shows the fitted values from the kernel regression for the sample of segments that are above the firm's average q and above the firm's asset-weighted opportunities. The bottom plot shows the fitted values from the kernel regression for the sample of segments that are below the firm's average q and below the firm's asset-weighted opportunities.

Table V
Value Added by Allocation and Diversity
in Investment Opportunities

This table estimates the empirical link between different measures of the efficiency of the investment policy of a diversified firm and diversity across segments. In the first column the dependent variable is the relative value added by allocation, defined as

$$\frac{\sum_{j=1}^n BA_j(q_j - \bar{q}) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right) \right)}{BA}$$

where $\bar{q}$ is the asset-weighted average of segment q s for the firm, q_j is the asset-weighted q ratio of single-segment firms that operates exclusively in segment j , I_j is the capital expenditure of segment j (item #4 of the COMPUSTAT segment file), BA_j is the book value of assets of segment j , and (I_j^s/BA_j^s) is the asset-weighted average capital expenditure to assets ratio for single-segment firms in the corresponding industry. The dependent variable in the second column is the absolute value added by allocation, defined as

$$\frac{\sum_{j=1}^n BA_j(q_j - 1) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}$$

The inverse of q equals $1/qe$, where qe is the equally weighted average q over segments in the firm. Diversity is the standard deviation of a firm's asset-weighted q ($\sqrt{\sum_{j=1}^n [(w_j q_j - \bar{w} \bar{q})^2 / (n - 1)]}$) divided by the equally weighted average q . Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2
Inverse of average q	0.007 (7.532)	-0.004 (-1.439)
Diversity	-0.008 (-5.543)	-0.010 (-2.372)
Firm size	0.000 (-0.059)	0.004 (3.220)
R^2	0.325	0.394
N	13,946	13,946

Table VI
Excess Value and Efficiency of Investments

This table estimates the relation between excess value of a diversified firm and efficiency of investments. The dependent variable is the excess value of a diversified firm vis-à-vis a single-segment firm. In the first two columns this is measured as

$$EV = \frac{MV}{RVA} - \sum_{j=1}^n q_j \frac{BA_j}{BA},$$

where MV is the market value of assets, RVA the replacement value of the assets, BA the book value of assets, subscript j refers to segment j , n is the total number of segments, and q_j is the asset-weighted average Tobin's q of single-segment firms that operate in the three-digit industry of segment j . In the last two columns this is measured using market-to-sales as

$$EV' = \frac{MV}{S} - \sum_{j=1}^n \left(\frac{MV}{S} \right)_j \frac{S_j}{S},$$

where MV is the market value of assets, S is the value of sales, n is the number of segments in the diversified firm, $(MV/S)_j$ is the sales-weighted average market-to-sales ratio of single-segment firms in the same three-digit industry, and subscript j refers to segment j . The relative value added by allocation is

$$\frac{\sum_{j=1}^n BA_j(q_j - \bar{q}) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right) \right)}{BA},$$

where $\bar{q}$ is the asset-weighted average of segment q s for the firm, q_j is the asset weighted q ratio of single-segment firms that operates exclusively in segment j , I_j is capital expenditure of segment j (item #4 of the COMPUSTAT segment file), BA_j is the book value of assets of segment j , and (I_j^s/BA_j^s) is the asset-weighted average capital expenditure to assets ratio for single-segment firms in the corresponding industry. The absolute value added by allocation is

$$\frac{\sum_{j=1}^n BA_j(q_j - 1) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}.$$

Standard deviation of segment q s is the standard deviation of the asset-weighted q s of the segments in which the firm operates. Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2	3	4
Relative value added by allocation	1.077 (2.674)		0.995 (2.171)	
Absolute value added by allocation		0.814 (4.943)		1.330 (6.592)
Firm size	0.022 (1.216)	0.019 (1.013)	-0.210 (-7.902)	-0.217 (-8.149)
R^2	0.633	0.634	0.698	0.701
N	13,868	13,868	12,169	12,169

5.8 Table VI: Better allocation is associated with higher firm value

Read.

- Excess value rises with the relative value-added-by-allocation measure; the coefficient is about **1.077** using the q -based excess-value measure and about **0.995** using the market-to-sales measure.
- Excess value also rises with the absolute value-added-by-allocation measure; the coefficient is about **0.814** with the q -based value metric and about **1.330** with the market-to-sales metric.

Economic message. The paper is not only showing that transfers move in a theory-consistent way. It also shows that *the direction of those transfers matters for value*. Allocating relatively more to better-opportunity segments raises firm value.

Table VII
Excess Value of a Diversified Firm and Diversity

This table estimates the relation between excess value of a diversified firm and the diversity of investment opportunities in its segments. The dependent variable is the excess value of a diversified firm vis-à-vis a portfolio of single-segment firms in the same three-digit industries. In the first two columns, excess value is

$$EV = \frac{MV}{RVA} - \sum_{j=1}^n q_j \frac{BA_j}{BA},$$

where MV is the market value of assets, RVA is the replacement value of the assets, n is the number of segments in the diversified firm, BA is the book value of assets of the whole firm, subscript j refers to segment j . In the last two columns, excess value is

$$EV' = \frac{MV}{S} - \sum_{j=1}^n \left(\frac{MV}{S} \right)_j \frac{S_j}{S},$$

where MV is the market value of assets, S is the value of sales, n is the number of segments in the diversified firm, $(MV/S)_j$ is the sales-weighted average market-to-sales ratio of single-segment firms in the same three-digit industry, subscript j refers to segment j . The inverse of equally weighted q equals $1/q$, where q is the equally weighted average q over segments in firm. Diversity is the standard deviation of the segments' asset-weighted q ($\sqrt{\sum_{j=1}^n [(w_j q_j - \bar{w} \bar{q})^2 / (n - 1)]}$) divided by the equally weighted average q . The Herfindahl index of segment's size is based on the assets of the segment. Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2	3	4
Inverse of average q	0.420 (15.078)	0.421 (15.095)	0.179 (4.978)	0.180 (4.983)
Diversity	-0.276 (-5.686)	-0.367 (-6.412)	-0.169 (-2.892)	-0.291 (-4.184)
Herfindahl index of division size		0.214 (3.049)		0.280 (3.320)
Company's size	0.025 (1.385)	0.033 (1.766)	-0.209 (-7.938)	-0.200 (-7.546)
R^2	0.643	0.643	0.700	0.700
N	13,868	13,868	12,169	12,169

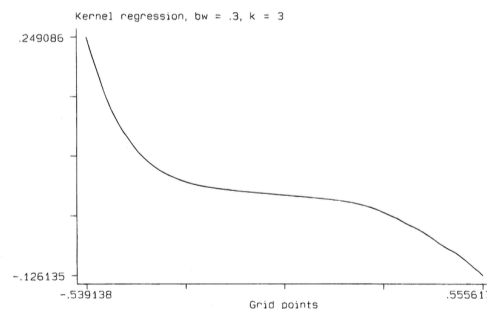


Figure 4. Kernel regression of the excess value on diversity. We plot the fitted values from the kernel regression of the relation between excess value and diversity. Before estimating the relationship, we partial out the inverse of the equally weighted q , firm size (logarithm of total sales), firm-specific effects as well as the calendar-year dummies. The kernel density is estimated using the Epanechnikov kernel with a bandwidth of 0.5 and a grid of 100 points. The y-axis contains the fitted values from the kernel regression and the x-axis the corresponding grid points.

5.9 Table VII and Figure 4: Diversity directly lowers excess value

Read.

- In the direct reduced-form regressions, the diversity coefficient is about **-0.276** without the Herfindahl control and about **-0.367** with the Herfindahl control when excess value is measured using the q -based metric.
- Using the market-to-sales excess-value metric, the coefficients are about **-0.169** and **-0.291**.
- The kernel plot in Figure 4 shows a strongly negative nonparametric relation between excess value and diversity.

Economic message. The discount is not just a by-product of one specific allocation metric. Even

in reduced form, diverse conglomerates look less valuable.

6 Short summary

- Diversified firms can create value by reallocating funds, but they also create internal political distortions.
- The relevant state variable is not raw segment q alone but **resource-weighted opportunities**.
- When diversity in weighted opportunities rises, headquarters increasingly uses transfers to manage incentives rather than to maximize static allocative efficiency.
- Those transfers often go toward relatively weak segments.
- The resulting allocation pattern reduces the value added by internal capital allocation and lowers excess firm value.
- The paper's most distinctive contribution is to show that **who gets subsidized depends on weighted opportunities, not simply on raw opportunities**.

7 Conclusion

7.1 Core conclusion

The paper gives a unified theory and empirical test of why diversification can be costly. Internal capital markets are not simply miniature versions of efficient external capital markets. Because divisions bargain over surplus after investments are chosen, headquarters may use transfers to manipulate bargaining positions rather than to put every dollar where its static return is highest.

7.2 Economic intuition

The strongest division is exactly the one most tempted to protect itself rather than cooperate. If headquarters wants that division to keep making efficient investments, it may have to raise the contribution of weaker divisions by giving them more resources ex ante. This is why diversity can generate cross-subsidies that look irrational from a pure net-present-value perspective but make sense as a second-best response to internal politics.

7.3 Takeaway

The diversification discount is not just about spanning many industries. It is about how **far apart** a firm's segments are in size and opportunities. Firms with low diversity can still benefit from internal markets. Firms with high diversity are more likely to misallocate capital and destroy value.